

Do Not Underestimate Lyssavirus – Amy Barnes

Lyssavirus rabies is the Latin name for the rabies virus, which continues to frustrate efforts at eradication with features that make it hard to detect. The virus remains firmly entrenched within global wildlife populations from ancient times into modern years, even after other viruses have been declared eradicated. An infection brings terrifying symptoms, the brutality of which mean that fatality is a relief and not just the end of life. This virus should never be underestimated.

Lyssavirus should be considered “clever” in that it can pull off some amazing feats for a virus: a super-long incubation period, an ability to hide from immune systems, and the ability to use common mammals as potential hosts. The incubation period is typically advertised as being short, but there are outlying cases that reveal a longer potential timeframe, with the potential to propagate this timeframe as the virus spreads. Authors T.A. Boland and D. McGuone, in “Phylogenetic and epidemiologic evidence of multiyear incubation in human rabies,” reveal that Lyssavirus has an incubation period that can extend to nearly a decade: “Viral sequencing data provided confirmation that rabies encephalomyelitis may present after a long, multiyear incubation period, a consideration that previously has been hypothesized without the ability to exclude a more recent exposure.” Rabies has had an ace up its metaphorical sleeve: it can hide from immune systems using the host’s chemistry and cellular death to block awareness of its existence. In their article “Lyssaviruses and the Fatal Encephalitic Disease Rabies,” authors TP Scott and LH Nel explain:

“Pathogenic rabies lyssavirus strains inhibit innate immune signaling and induce cellular apoptosis as the virus progresses to the central nervous system and brain using viral protein facilitated retrograde axonal transport.” This hijacking of immunity has enabled animals to remain asymptomatic long enough to facilitate transmission across its hosts, which are mammals. The variety of hosts is wide enough to make it extremely difficult to fight rabies: “Rabies lyssavirus (RABV), is thought to be able to infect all terrestrial mammals.” (qtd. in Scott) This super-wide compatibility with potential hosts nearly guarantees its persistence. Finally, the effort to eradicate this virus has expanded in scope to cover thousands of species, to the point that this task can seem insurmountable.

Lyssavirus rabies’ reach extends most within untreated, unchecked, and ungoverned wildlife that crosses paths with every human population in the world, the most hazardous being dogs. Dogs remain a particular focus, in part, due to their lengthy relationship with humans and the access that humans readily grant them to their lives and living spaces. In Arnaud Tarntola’s “Four Thousand Years of Concepts Relating to Rabies in Animals and Humans, Its Prevention and Its Cure,” medical and legal texts from nearly four thousand years ago, extending into the middle ages and modern times, are described as containing much of the same basic understandings – and futility – that the humans of today have encountered. Tarantula begins in ancient times: “Humans have lived alongside domesticated dogs for 14,000 years at least . . . They have also long been familiar with their diseases, which became more prevalent as populations and their

animals congregated in the cities that arose in Mesopotamia. Two cuneiform tablets . . . describe Sumerian rules and regulations attesting to the fact that a causal link between the bite of a rabid animal and a human death from rabies was well recognized almost 4000 years ago.” This link between dogs, the virus, and humans, exists worldwide. “Rabies is mentioned in many ancient texts, from the Vedic period (in ancient India ca. 1750–500 BCE), to ancient China, Egypt, and the Middle East as well as Greece and Rome.” (qtd. in Tarntola) The futility of attempts to treat patients was expressed in texts that included one final step taken after all treatments failed, in Primelin, France: “at Saint-Tügen chapel, patients with declared rabies were stifled between mattresses.” (qtd. in Tarntola) In modern times, the virus continues to have a devastating impact. “An estimated total of 59,000 humans die of rabies each year, more than twice the estimated 28,600 deaths caused by the tragic 2014–2016 Ebola outbreak in West Africa.” (qtd. in Tarntola)

The symptoms of rabies in its classical form are similar to the portrayal of zombies: drooling, choking, inability to swallow, and aggression. The “dumb” form of the disease is characterized by paralysis and numbness. Terence Peter Scott and Louis Hendrik Nel provide further clarification regarding how encephalitic (furious or classical) is not like paralytic (dumb) rabies: “The encephalitic form of rabies is more common and presents in approximately 80% of patients, of which between 50 – 80% present with the classic symptoms such as hydrophobia and aerophobia – symptoms that are unique to rabies.” (qtd. in

Scott) Strain-specific symptoms vary: “Bat RABV infections more commonly present with tremors and involuntary twitching/jerking (myoclonus), while dog strains more frequently present with classical hydrophobia and aerophobia.” (qtd. in Scott) While rabies is fatal, the causes of death differ by strain as well: “Encephalitic rabies typically progresses to severe flaccid paralysis, coma and death caused by multiple organ failure, in contrast to paralytic rabies which manifests with prominent muscle weakness early in the course of illness.” (qtd. in Scott) The grim outcome does share one commonality: severe brain damage. One autopsy of a victim of the virus revealed how complete the damage to the brain can be: “The brain was dusky gray and liquefied with obliteration of all anatomic landmarks.” (qtd. in Boland) The progression towards death is torture for victims, which leads to a question of how not to fall prey to Lyssavirus rabies.

The Centers for Disease Control, in “Rabies Prevention and Control,” warns: “You must get medical help right away. This is called post-exposure prophylaxis (PEP), and it's highly effective if you get it as soon as possible after exposure.” This warning is applicable across all forms of contact: bites, scratches, and contact. The message is clear: do not underestimate Lyssavirus rabies.

Annotated Bibliography – Amy Barnes

Thesis: Do not underestimate Lyssavirus.

T.A. Boland, D. McGuone, et al. “Phylogenetic and epidemiologic evidence of multiyear incubation in human rabies.” *Annals of Neurology*, vol. 75, no. 1, 2014, pp. 155-160. *PubMed*, <https://doi.org/10.1002/ana.24016>

In “Phylogenetic and epidemiologic evidence of multiyear incubation in human rabies,” T. A. Boland and D. McGuone warn of lengthy incubation periods via an example medical case. Intended for medical personnel and researchers encountering Lyssavirus, its production is designed to cure ignorance regarding a potential for incubation periods far longer than the mainstream days-to-months.

TP Scott TP, LH Nel. “Lyssaviruses and the Fatal Encephalitic Disease Rabies.” *Frontiers in Immunology*, vol. 12, 2021, doi: 10.3389/fimmu.2021.786953

Researchers TP Scott and LH Nel provide a comprehensive overview of how Lyssavirus Rabies spreads, functions, and persists in their article “Lyssaviruses and the Fatal Encephalitic Disease Rabies.” They write for a general audience of researchers, medical students, and the interested public to raise awareness regarding a need for eradication efforts to continue, as success in some countries becoming impetus to demand a continued fight to wipe the deadliest disease off the planet.

Tarantola, Arnaud. “Four Thousand Years of Concepts Relating to Rabies in Animals and Humans, Its Prevention and Its Cure.” *Tropical Medicine and Infectious Diseases*, Special Issue “Rabies Symptoms, Diagnosis, Prophylaxis and Treatment” vol. 2(2), No. 5, 2017, *PubMed*, <https://doi.org/10.3390/tropicalmed2020005>

Arnaud Tarantola explores the global history of frustrated humans fighting rabies in “Four Thousand Years of Concepts Relating to Rabies in Animals and Humans, Its Prevention and Its Cure.” A special issue of *Tropical Medicine and Infectious Diseases*, dedicated to rabies, includes this article to call for vaccine and immunoglobulin treatment availability to populations at risk of infection.

U.S. Department of Health and Human Services, Centers for Disease Control and Prevention. “Rabies Prevention and Control.” CDC, 1 July 2025, <https://www.cdc.gov/rabies/prevention/index.html>.

The CDC lays out a public-facing summary including treatments in “Rabies Prevention and Control.” The general public is the intended audience, and the goal is to meet the need for immediate access to information regarding treatment of suspected exposure to Lyssavirus rabies.

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Outline

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- IV. Persistence Beyond Smallpox
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